MILITARY SPECIFICATION

TUBE, STOMACH, SURGICAL, PLASTIC RADIOPAQUE, LEVIN

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers non-sterile Levin stomach tubes suitable for use in aspirating, tube feeding and irrigating.

1.2 Classification. Tubes covered by this specification shall be of the following sizes, as specified (see 6.1).

Size 12	-	12 French (Fr.)
Size 16	-	16 French (Fr.)
Size 18	-	18 French (Fr.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive Water-Resistant (for sealing fiberboard boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for carton sealing).

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

23154	Tube, Stomach, Surgical, Plastic, Radiopaque, Levin

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.
The Pharmacopeia of the United States of America (U.S.P.)

(Applications for copies shall be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042).

AMERICAN SOCIETY FOR TESTING AND MATERIALS
D-412 Tension Testing for Vulcanized Rubber.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103).

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3. REQUIREMENTS

3.1 Material. Tubes shall be fabricated of flexible, transparent, medical grade polyvinyl plastic, meeting the requirements of class II plastics of the current edition of the Pharmacopeia of the United States (USP), when tested as specified in 4.4.1. Material shall not contain more than five parts per million (p.p.m.) of lead when tested as specified in 4.4.2. Material shall not be affected by oil base lubricants.

3.2 Style, design, and dimensions. Shall be a clean single lumen stomach tube of the Levin design, with integral tapered connector at the proximal end. Shall have a distal hole and four oval holes equally spaced, two aligned on a radiopaque stripe and two on the diametrically opposite side. All holes shall have well rounded, softly beveled edges. Tubes shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 23154.

3.3 Physical properties.

3.3.1 Breaking strength. The breaking strength of the tubes shall be as specified in Table I for the appropriate size, when tested as specified in 4.4.3.

Table I. Breaking force

Size	12	16	18
Breaking strength (lb.) min.	12.0	17.5	25.0

3.4 Sterilization. Tubes shall be capable of being sterilized by commercially prepared cold sterilization solutions and by steam autoclaving. When tested as specified in 4.4.4, tubes shall not become distorted, cracked, or adversely affected so as to impair their use.

3.5 Identification marking. Each tube shall be permanently and legibly marked with the manufacturer's name or registered trademark, and the nominal French size.

3.6 Instructions. The label on or within the immediate container shall bear complete instructions for use, and in addition, shall include the following or similar statement: "NOTICE: This item takes on a milky-translucent appearance after steam sterilization. This change has no deleterious effect on the serviceability of the item."

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3.7 Manufacturer's record. The manufacturer shall place a lot control number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made available for inspection by any duly qualified medical officer of the Armed Forces, National Institutes of Health, or any official of the Federal Food and Drug Administration. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.8 Workmanship. Workmanship shall be first class in every respect. Tubes shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastic, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. The unit of product for examination shall be one tube.

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Table II. Sampling for Examination

Defect category	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table III.

Table III. Sampling for Tests

Component (unit of product)	Characteristic	Requirement	Test procedure	Insp. level	AQL (percent defective)
1-Tube	Material-Class II plastic	3.1	4.4.1	**	*
	Lead content	3.1	4.4.2	S-2	*
	Breaking strength	3.3.1	4.4.3	S-2	*
	Sterilization	3.4	4.4.4	S-2	*
	Dimensions	3.2	---	S-1	*

* Acceptance number shall be zero for all sample sizes.
** Sufficient samples to perform the biological tests shall be selected from each lot.

4.2.3 Raw material. Ten grams of each lot of plastic used in the fabrication of the tubes shall be selected for the lead test.

4.2.4 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

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4.3 Examination. The tubes shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table IV. Inspection, however, shall not be restricted to the classified possible defects listed in Table IV.

Table IV. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Tube not symmetrical in shape.
102	Surface of tube not smoothly finished.
103	Tube not free of bloom or tackiness.
104	Inside of tube not free of obstructing material.
105	Holes not spaced and aligned as specified.
106	Holes not smooth and free of flash.
107	Tube not marked with French size.
Minor	
201	Tube not free of discoloration.
202	Tube not free of pits, blisters, scratches, or crazing.
203	Tube holes not symmetrical.
204	Packaging, Packing and marking not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Biological tests. The tubes shall be tested in accordance with the requirements of the current edition of the U.S.P. for class II plastic.

4.4.2 Test for lead. Ten grams of plastic shall be taken, ignited to char, and then further ignited in a muffler at 700° C, to a white ash. The ash shall be boiled in 15 milliliters (ml.) of 10 percent by volume of concentrated nitric acid (specific gravity of 1.4, chemically pure or analytical reagent grade), filtered into a volumetric flask and made up to 100 ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the current edition of the Pharmacopeia of the United States, shall be applied using a control of 5 p.p.m. or 12 p.p.m. of lead solution as given in the U.S.P. test. If the sample solution is darker than the limit, smaller aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in p.p.m. is determined.

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4.4.3 Physical properties tests. The breaking strength of the tubes shall be tested in accordance with ASTM Method D412 at a speed of separation of 20 ± 2 inches per minute.

4.4.4 Sterilization.

4.4.4.1 Steam sterilization. Tubes shall be steam autoclaved for 20 minutes at 250 degrees Fahrenheit (°F.) and 15 pounds per square inch gage (psig) than allowed to cool to room temperature (68° F. to 72° F.).

4.4.4.2 Cold sterilization. Tubes shall be immersed in a chemical disinfectant for 20 minutes at room temperature (68° F. to 72° F.), and then followed by five water rinses.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package (50 tubes), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	6 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Individual package. Each tube shall be individually packaged in a heat seal, transparent, plastic envelope.

5.1.3.2 Unit package. Each unit (50 tubes) shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the box specification.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual package. In additon to any commercial markings, each individual package shall be marked with the item identification, size, length and the legend "NOT STERILE". Marking shall be printed directly on the bag or on a label inserted within the bag in such a manner as to permit ready identification.

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5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Sizes required (see 1.2).
c. Levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification is intended to cover tubes normally procured to meet military requirements. It does not include all types or sizes as indicated by the title, nor all types commercially available.

6.3 This specification covers the following items appearing in the Federal Supply Catalog:

Stock Number	Item Identification	Size
6515-225-9719	TUBE, STOMACH, SURGICAL, Plastic, Radiopaque, Levin, 16 Fr, 50s	16
6515-935-4063	TUBE, STOMACH, SURGICAL, Plastic, Radiopaque, Levin, 12 Fr, 50s	12
6515-935-4065	TUBE, STOMACH, SURGICAL, Plastic, Radiopaque, Levin, 18 Fr, 50s	18

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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